

Problem 11.19

You are given

$$\delta_t = \frac{2}{t-1}, \quad 2 \leq t \leq 20.$$

For any one-year interval between n and $n+1$, with $2 \leq n \leq 9$, calculate $d_{n+1}^{(2)}$.

Solution

The discount factor over the interval from time n to time $n+1$ is

$$\begin{aligned} v_{n,n+1} &= \exp \left(- \int_n^{n+1} \delta_t dt \right) \\ &= \exp \left(- \int_n^{n+1} \frac{2}{t-1} dt \right) \\ &= \exp \left(-2 \ln(t-1) \Big|_n^{n+1} \right) \\ &= \exp \left[-2(\ln n - \ln(n-1)) \right] \\ &= \exp \left(\ln \left(\frac{n-1}{n} \right)^2 \right) \\ &= \left(\frac{n-1}{n} \right)^2. \end{aligned}$$

For a nominal annual discount rate convertible twice per year, the corresponding one-year discount factor is

$$\left(1 - \frac{d_{n+1}^{(2)}}{2} \right)^2.$$

Therefore,

$$\begin{aligned} \left(1 - \frac{d_{n+1}^{(2)}}{2} \right)^2 &= \left(\frac{n-1}{n} \right)^2, \\ 1 - \frac{d_{n+1}^{(2)}}{2} &= \frac{n-1}{n}, \\ d_{n+1}^{(2)} &= 2 \left(1 - \frac{n-1}{n} \right). \end{aligned}$$

Thus,

$$\boxed{d_{n+1}^{(2)} = \frac{2}{n}}, \quad 2 \leq n \leq 9.$$